

0.1 Section 4.1 - Divisibility

Question

When is an integer divisible by 2?

Solution:

When an integer ends in 0, 2, 4, 6, 8

Let's prove this after we develop some notation

Definition 1 (Divides)

For integers $a, b \in \mathbb{Z}$, we say "a divides b" if b is divisibly by a and we write

$$a \mid b \quad (\text{TRUE value})$$

$$a \nmid b \quad (\text{FALSE value})$$

Example 1

$2 \mid 48$ since 48 is divisible by 2

but $7 \nmid 48$ since 48 is not divisible by 7

Let's prove

for $n \in \mathbb{Z}$, $2 \mid n$ if $\exists k \in \mathbb{Z}$ such that $n = 10k$ or $10k + 2$ or $10k + 4$ or $10k + 6$ or $10k + 8$

We could also say

$\exists k \in \mathbb{Z}$ such that $n = 10k + r$ where

$$r = \{0, 2, 4, 6, 8\}$$

Proof

(Proof of the converse)

Let $n \in \text{set } \mathbb{Z}$

Then base 10 representation of n is of the form

$$n = d_k 10^k + d_{k-1} 10^{k-1} + \dots + d_1 10 + d_0$$

Since $2 \mid n$, then $n = 2m$ for some $m \in \mathbb{Z}$

We have

$$\begin{aligned}n &= 10(d_k 10^{k-1} + d_{k-1} 10^{k-2} + \dots + d_1) + d_0 \\n &= 2 \left[5(d_k 10^{k-1} + d_{k-1} 10^{k-2} + \dots + d_1) \right] + d_0\end{aligned}$$

Solving for d_0 we get

$$d_0 = n - 2 \left[5(d_k 10^{k-1} + d_{k-1} 10^{k-2} + \dots + d_1) \right]$$

Since d_0 is the sum of two even numbers, then d_0 is even and $d_0 \in \{0, 2, 4, 6, 8\}$

Let's prove some of the properties of the "divides" symbol

In particular, let's rewrite what $a \mid b$ means for $a, b \in \mathbb{Z}$

So, $a \mid b$ means

$$\exists k \in \mathbb{Z} \text{ such that } b = ak$$

We believe if two integers are divisible by an integer, then their sum is divisible by that integer

We prove $a, b, c \in \mathbb{Z}, a \mid b, a \mid c, \implies a \mid (b + c)$

Proof

$$a \mid b \implies \exists k \in \mathbb{Z} \text{ such that } b = ak$$

$$a \mid c \implies \exists n \in \mathbb{Z} \text{ such that } c = an$$

Then

$$b + c = ak + an = a(k + n)$$

$$\therefore a \mid (b + c) \text{ as desired}$$

Note

This also works for **subtraction**

Proposition 1 (Modulus Multiplication)

$a, b, c \in \mathbb{Z}, a \mid b, a \mid c \implies a^2 \mid bc$

Proof

$$a \mid b \implies \exists k \in \mathbb{Z} \text{ such that } b = ak$$

$$a \mid c \implies \exists n \in \mathbb{Z} \text{ such that } c = an$$

Then

$$\begin{aligned} bc &= (ak)(an) \\ &= a^2(kn), \text{ proving } a^2 \mid bc \text{ as desired} \end{aligned}$$

Proposition 2 (Constant Factor)

$a, b, c \in \mathbb{Z}, a \mid b \implies a \mid bc$

Proof

$$a \mid b \implies \exists k \in \mathbb{Z} \text{ such that } b = ak$$

$$bc = akc = a(kc), \text{ implying } a \mid bc, \text{ as desired}$$

Let's rewrite the old proof from earlier, this time proving the actual proposition rather than contrapositive

Proposition 3

For $n \in \mathbb{Z}$, where $n = 10k + r, r \in \{0, 2, 4, 6, 8\}$, then $2 \mid n$

Proof

$$\forall r \in \{0, 2, 4, 6, 8\}, \exists m \in \mathbb{Z} \text{ such that } r = 2m$$

Subbing this in

$$\begin{aligned}
 n &= 10k + r = 10k + 2m \\
 &= 2(5k + m), \text{ implying } 2 \mid n, \text{ as desired}
 \end{aligned}$$

Proposition 4

For $n \in \mathbb{Z}$, if the sum of the digits is divisible by 3, then the integer n is divisible by 3

Proof.

$$\begin{aligned}
 n &= d_k 10^k + d_{k-1} 10^{k-1} + \dots + d_1 10 + d_0 \\
 n &= (10^k - 1)d_k + (10^{k-1} - 1)d_{k-1} + \dots + 99d_2 + 9d_1 + \underbrace{(d_k + d_{k-1} + \dots + d_2 + d_1 + d_0)}_{\text{sum of digits}}
 \end{aligned}$$

Because 10^k and so on will always be a bunch of 9's, we see that

$$3 \mid (10^k - 1)d_k + (10^{k-1} - 1)d_{k-1} + \dots + 99d_2 + 9d_1$$

And

$$3 \mid (d_k + d_{k-1} + \dots + d_2 + d_1 + d_0) \text{ by hypothesis}$$

$\therefore n$ is the sum of 2 multiples of 3, proving that $3 \mid n$ as desired

□

Here's just a table for divisibility rules

n is divisible by	if
2	last digit is even
3	sum of digits is divisible by 3
4	last 2 digits is divisible by 4
5	last digit is 0 or 5
6	divisible by both 2 and 3
7	He showed us but I forgot
8	last 3 digits divisible by 8
9	sum of digits is divisible by 9
10	last digit is 0

Definition 2 (Division Algorithm)

The **division algorithm** is a way of expressing division with integers

$$\forall a \in \mathbb{Z} \forall d \in \mathbb{Z}^+ \exists q \in \mathbb{Z} \exists r \in \{0, 1, 2, 3, \dots, d-1\}, a = dq + r$$

Where

d : divisor

q : quotient

r : remainder

Example 2

Let $a = 341$ and $d = 7$

$$341 = \underbrace{7}_{\text{divisor}} \cdot \underbrace{48}_{\text{quotient}} + \underbrace{5}_{\text{remainder}}$$

However we can write this in a more compact way

$$341 \pmod{\underbrace{7}_{\text{divisor}}} = \underbrace{5}_{\text{remainder}}$$

Example 3

Let's give examples of division, we're practically finding the remainder

1.

$$\frac{6568}{9} = 729.777$$
$$6568 - 9 \cdot 729 = 7$$

2.

$$\frac{343348}{7} = 49049.714$$
$$343348 - 7 \cdot 49049 = 5$$

3.

$$\begin{aligned}\frac{-100}{13} &= -7.6923 \\ -100 - 13(-7) &= -9 \\ &= -100 - 13(-8) = 4 \\ -100 &= (13)(-8) + 4\end{aligned}$$

Note

For negatives, round down to the next negative integer in order to get a positive remainder

Example 4

Let's do more examples but with mod

1.

$$1023 \pmod{11} = 0$$

2.

$$6568 \pmod{9} = 7$$

3.

$$343348 \pmod{7} = 5$$

4.

$$-100 \pmod{13} = 4$$

Definition 3 (Congruence)

Let $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$

We say a is **congruent** to b modulo m if

$$a \pmod{m} = b \pmod{m}$$

This is basically saying are the remainders the same

Written as

$$a \equiv b \pmod{m}$$

And if they're not then

$$a \not\equiv b \pmod{m}$$

Note

If $a \equiv b \pmod{m}$ then by **Division Algorithm**

$$a = mq_a + r \text{ and } b = mq_b + r$$

Subtrating the equations we get

$$\begin{aligned} a - b &= (mq_a + r) - (mq_b + r) \\ &= m(q_a - q_b) \\ \therefore m &\mid (a - b) \end{aligned}$$

Proposition 5

Let $a, b, c, d \in \mathbb{Z}, m \in \mathbb{Z}^+$

Suppose $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$

1. $a + c \equiv b + d \pmod{m}$

Proof

$$\begin{aligned} a &\equiv b \pmod{m} \implies m \mid (a - b) \\ c &\equiv d \pmod{m} \implies m \mid (c - d) \end{aligned}$$

Then

$$\begin{aligned} m &\mid [(a - b) + (c - d)] \\ m &\mid [(a + c) - (b + d)] \\ \therefore a + c &\equiv b + d \pmod{m} \text{ as desired} \end{aligned}$$

2. $ac \equiv bd \pmod{m}$

Proof

$$\begin{aligned}a &\equiv b \pmod{m} \implies m \mid (a - b) \\c &\equiv d \pmod{m} \implies m \mid (c - d)\end{aligned}$$

$$\begin{aligned}m \mid (a - b) &\implies \exists k_1 \in \mathbb{Z} \text{ such that } a - b = mk_1 \\m \mid (c - d) &\implies \exists k_2 \in \mathbb{Z} \text{ such that } c - d = mk_2\end{aligned}$$

Then $a = b + mk_1$ and $c = d + mk_2$

$$\begin{aligned}ac &= (b + mk_1)(d + mk_2) && \text{(FOIL)} \\&= bd + bmk_2 + dm k_1 + m^2 k_1 k_2\end{aligned}$$

Moving bd

$$\begin{aligned}ac - bd &= bmk_2 + dm k_1 + m^2 k_1 k_2 \\&= m(bk_2 + dk_1 + mk_1 k_2) \\&\boxed{\therefore m \mid (ac - bd)}\end{aligned}$$

3. if $h \in \mathbb{Z}$ then $ah \equiv bh \pmod{m}$

Proof

$$a \equiv b \pmod{m} \implies m \mid (a - b)$$

Then since $h \in \mathbb{Z}$, we have

$$\begin{aligned}m &\mid h(a - b) \\m &\mid (ah - bh) \\&\boxed{ah \equiv bh \pmod{m}}\end{aligned}$$